พรีมภัทร ภาวัฒนวคุณ 673380594-9

Assignment 4 — TCP & UDP

Objectives

Part 1: Generate Network Traffic in Simulation Mode

Part 2: Examine the Functionality of the TCP and UDP Protocols

Part 1: Generate Network Traffic in Simulation Mode and View Multiplexing

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following task to reduce the amount of network traffic viewed in the simulation.

a. Click MultiServer and click the Desktop tab > Command Prompt.

b. Enter the ping -n 1 192.168.1.255 command. You are pinging the broadcast address for the client LAN. The command option will send only one ping request rather than the usual four. This will take a few seconds as every device on the network responds to the ping request from MultiServer.

c. Close the MultiServer window.

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping -n 1 192.168.1.255

Pinging 192.168.1.255 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=10ms TTL=128
Reply from 192.168.1.1: bytes=32 time=10ms TTL=128
Reply from 192.168.1.4: bytes=32 time=10ms TTL=128
Reply from 192.168.1.3: bytes=32 time=10ms TTL=128

Ping statistics for 192.168.1.255:
    Packets: Sent = 1, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 10ms, Average = 10ms
```

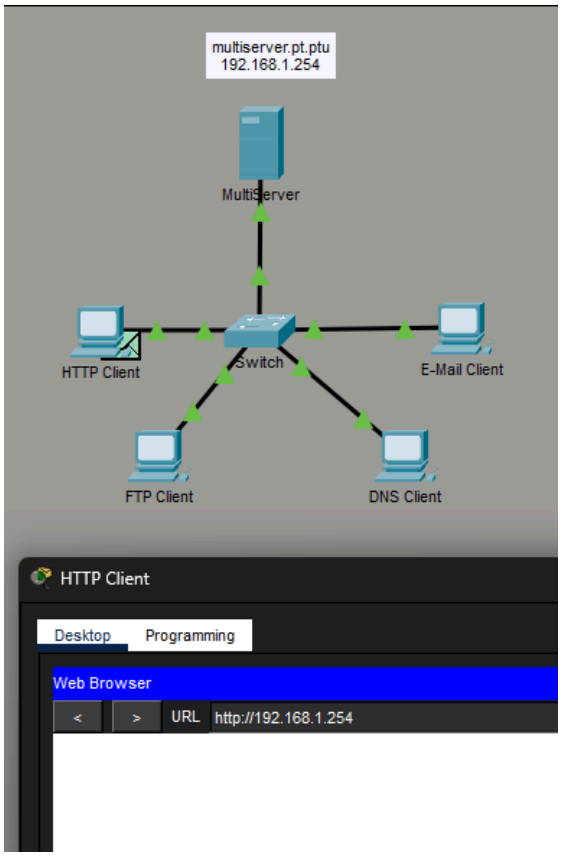
Step 2: Generate web (HTTP) traffic.

a. Switch to Simulation mode.

b. Click **HTTP Client** and open the **Web Browser** from the desktop.

c. In the URL field, enter **192.168.1.254** and click **Go**. Envelopes (PDUs) will appear in the topology window.

d. Minimize, but do not close, the **HTTP Client** configuration window.

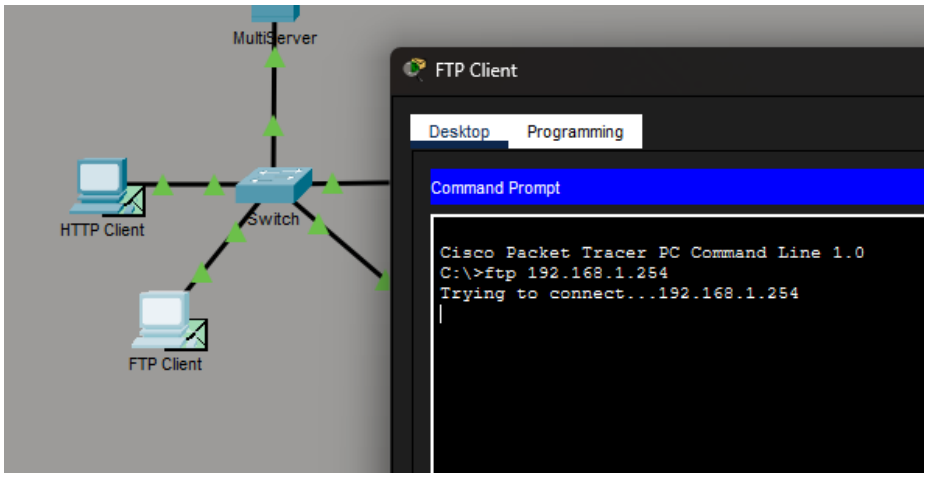


Step 3: Generate FTP traffic.

a. Click **FTP Client** and open the **Command Prompt** from the desktop

b. Enter the **ftp 192.168.1.254** command. PDUs will appear in the simulation window.

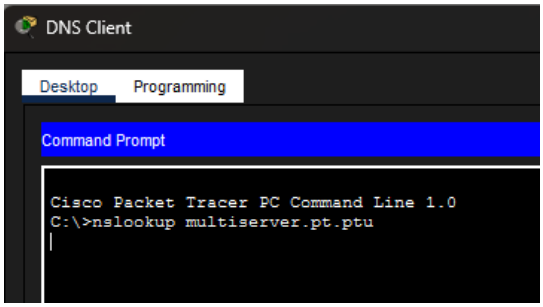
c. Minimize, but do not close, the **FTP Client** configuration window.

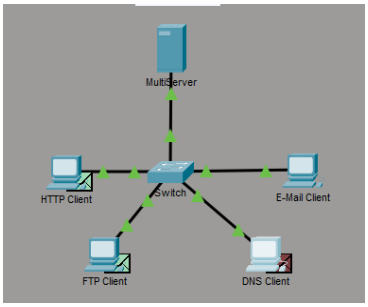


Step 4: Generate DNS traffic.

a. Click DNS Client and open the **Command Prompt**.

b. Enter the **nslookup multiserver.pt.ptu** command. A PDU will appear in the simulation window.





c. Minimize, but do not close, the **DNS Client** configuration window.

Step 5: Generate Email traffic.

a. Click **E-Mail Client** and open the **E Mail** tool from the Desktop.

b. Click **Compose** and enter the following information:

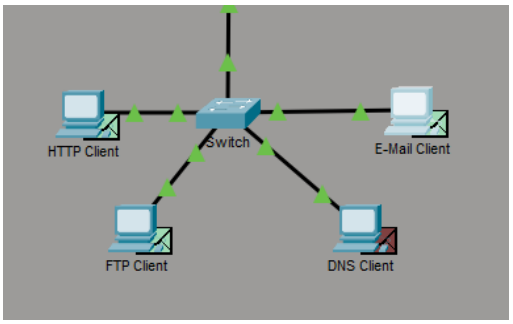
1) **To:** user@multiserver.pt.ptu

2) **Subject:** personalize the subject line

3) **E-Mail Body:** personalize the Email

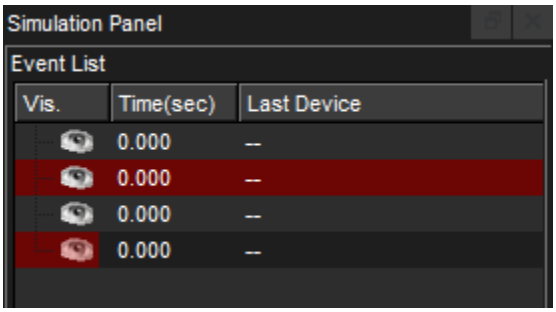
c. Click **Send**.

d. Minimize, but do not close, the **E-Mail Client** configuration window.



Step 6: Verify that the traffic is generated and ready for simulation.

There should now be PDU entries in the simulation panel for each of the client computers.



Step 7: Examine multiplexing as the traffic crosses the network.

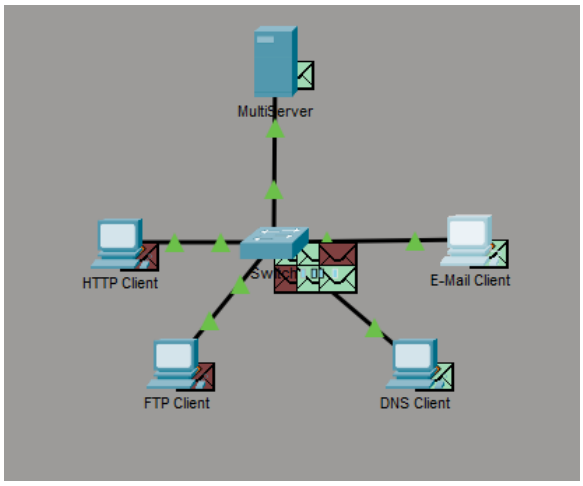
You will now use the **Capture/Forward button** in the Simulation Panel to observe the different protocols travelling on the network.

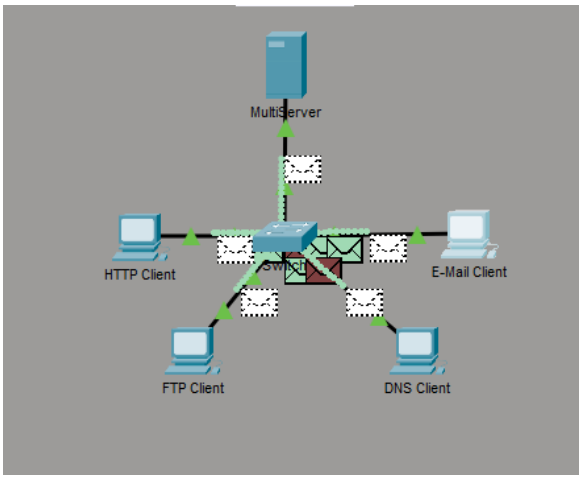
Note: The **Capture/Forward** button ‘ **>|** ‘ is a small arrow pointing to the right with a vertical bar next to it.

a. Click **Capture/Forward** once. All of the PDUs travel to the switch.

b. Click **Capture/Forward** six times and watch the PDUs from the different hosts as they travel on the network. Note that only one PDU can cross a wire in each direction at any given time.

What is this called?





A variety of PDUs appears in the event list in the Simulation Panel. What is the meaning of the different colors?

Question

Q: What is this called?
Ans :Multiplexing หรือก็คือหลาย protocol แชร์ network เดียวกัน

Q: สีต่างกันหมายถึงอะไร?

Ans : protocol ต่างชนิด (HTTP, DNS, FTP และอื่นๆ)

สีเขียว → ICMP / HTTP

สีฟ้า → DNS (UDP)

สีเหลือง → TCP

สีม่วง → ARP

สีแดง → Email / SMTP

Part 2: Examine Functionality of the TCP and UDP Protocols

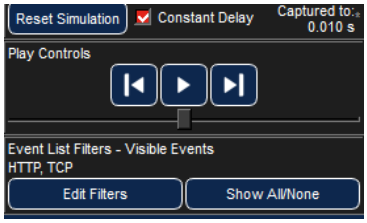
Step 1: Examine HTTP traffic as the clients communicate with the server.

a. Click **Reset Simulation**.

b. Filter the traffic that is currently displayed to only **HTTP** and **TCP** PDUs. To filter the traffic that is currently displayed:

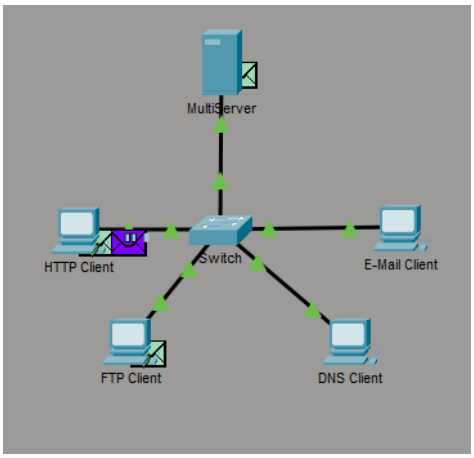
1) Click **Edit Filters** and toggle the **Show All/None** button.

2) Select **HTTP** and **TCP**. Click the red “x” in the upper right-hand corner of the Edit Filters box to close it. Visible Events should now display only **HTTP** and **TCP** PDUs.



c. Open the browser on HTTP Client and enter **192.168.1.254** in the URL field. Click **Go** to connect to the server over HTTP. Minimize the HTTP Client window.

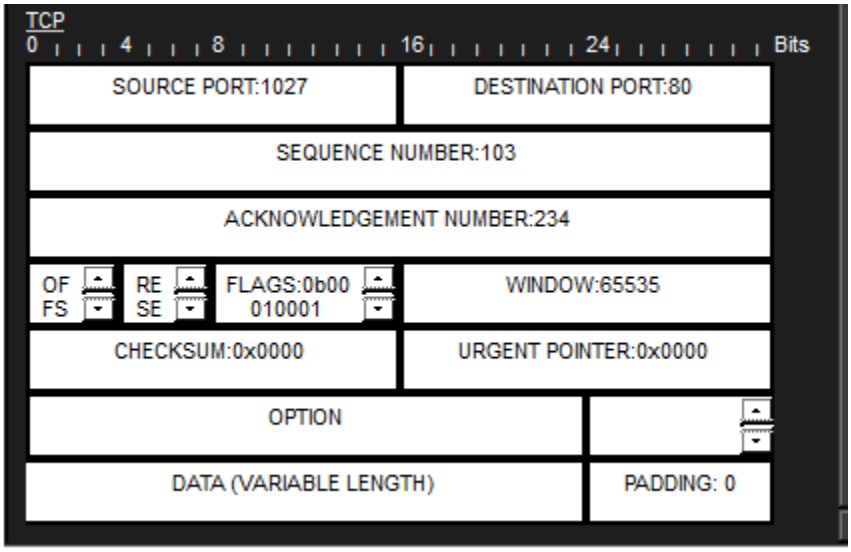
d. Click **Capture/Forward** until you see a PDU appear for HTTP. Note that the color of the envelope in the topology window matches the color code for the HTTP PDU in the Simulation Panel.



Question

Q: Why did it take so long for the HTTP PDU to appear?
Ans :เพราะ TCP ต้องทำการ three-way handshake ก่อนส่งข้อมูล HTTP จึงทำให้ HTTP PDU ปรากฏช้า

e. Click the PDU envelope to show the PDU details. Click the **Outbound PDU Details** tab and scroll down to the second to the last section.



Question

Q: What is the section labeled?
Ans :ส่วนดังกล่าวมีชื่อว่า TCP

Q: Are these communications considered to be reliable?
Ans :การสื่อสารนี้ถือว่ามีความน่าเชื่อถือ (Reliable) เนื่องจากใช้โปรโตคอล TCP ซึ่งมีการยืนยันการรับส่งข้อมูล

Q: Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values.
Ans :

SOURCE PORT : 1027

DESTINATION PORT) : 80

SEQUENCE NUMBER : 103

ACKNOWLEDGEMENT NUMBER : 234

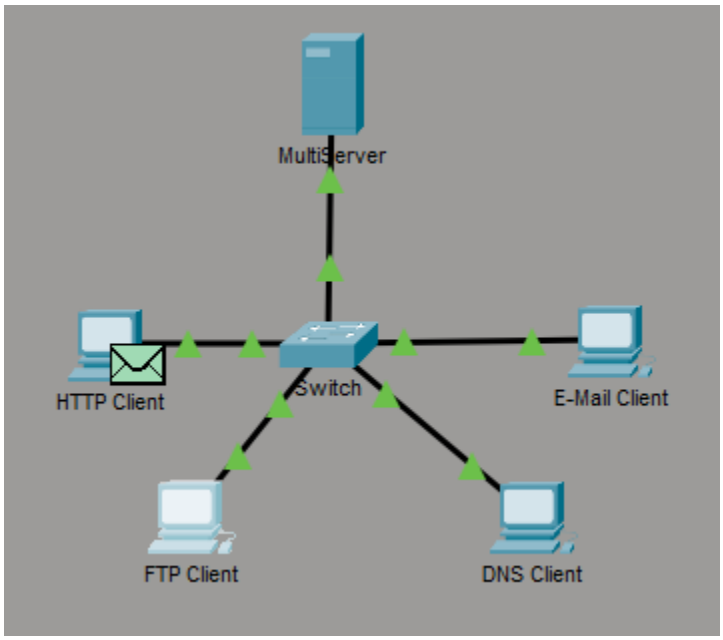
f. Look at the value in the Flags field, which is located next to the Window field. The values to the right of the “b” represent the TCP flags that are set for this stage of the data conversation. Each of the six places corresponds to a flag. The presence of a “1” in any place indicates that the flag is set. More than one flag can be set at a time. The values for the flags are shown below.

Flag Place	**6**	**5**	**4**	**3**	**2**	**1**
Value	URG	ACK	PSH	RST	SYN	FIN

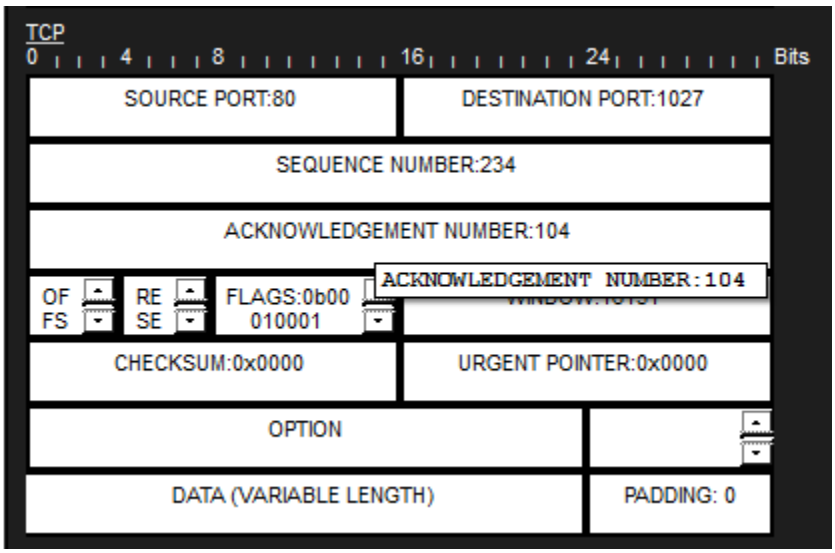
Question

Q: Which TCP flags are set in this PDU?
Ans :TCP flag ที่ถูกตั้งคือ SYN

g. Close the PDU and click **Capture/Forward** until a PDU with a checkmark returns to the **HTTP Client**.



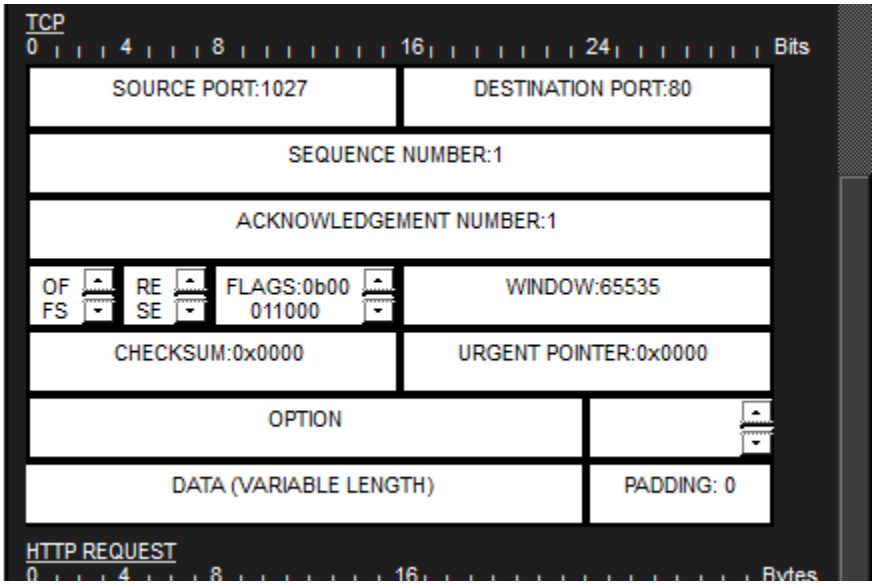
h. Click the PDU envelope and select **Inbound PDU Details**.



Question

Q: How are the port and sequence numbers different than before?
Ans :หมายเลขพอร์ตต้นทางและปลายทางถูกสลับกัน เนื่องจากเป็นการตอบกลับจากเซิร์ฟเวอร์ไปยังไคลเอนต์ และค่า sequence number กับ acknowledgement number มีการเพิ่มขึ้นตามลำดับของข้อมูลที่ส่ง

i. Click the HTTP PDU which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.



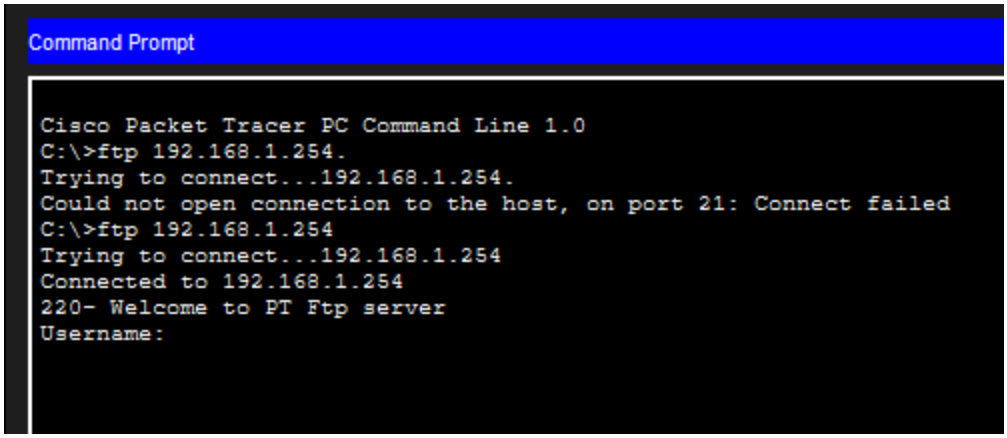
Question

Q: What information is now listed in the TCP section?
Ans :ในส่วน TCP แสดงข้อมูลเกี่ยวกับการรับส่งข้อมูลจริง เช่น sequence number, acknowledgement number และ flag ACK ซึ่งแสดงว่าการเชื่อมต่อถูกสร้างเรียบร้อยแล้วและเริ่มส่งข้อมูล HTTP

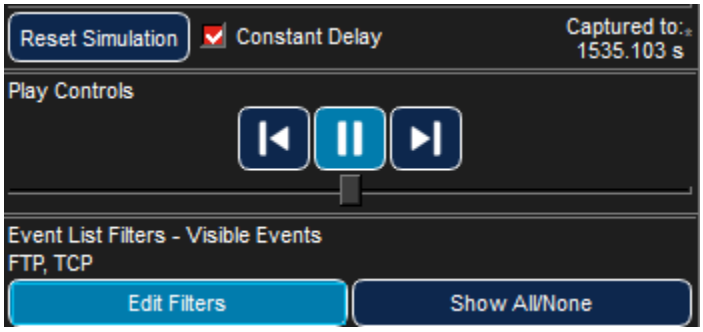
Q: How are the port and sequence numbers different from the previous two PDUs?
Ans :หมายเลขพอร์ตยังคงเดิม แต่ค่า sequence number และ acknowledgement number เปลี่ยนไปตามลำดับของการสื่อสารที่ดำเนินต่อไปหลังจากการสร้างการเชื่อมต่อสำเร็จ

Step 2: Examine FTP traffic as the clients communicate with the server.

a. Open the command prompt on the FTP Client desktop. Initiate an FTP connection by entering **ftp 192.168.1.254**.

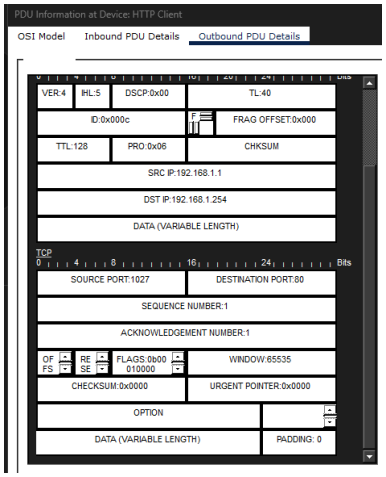


b. In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.



c. Click **Capture/Forward**. Click the second PDU envelope to open it.

Click the **Outbound PDU Details** tab and scroll down to the TCP section.



Question

Q: Are these communications considered to be reliable?
Ans :ถือว่าเป็นการสื่อสารที่เชื่อถือได้ เนื่องจาก FTP ใช้โปรโตคอล TCP ซึ่งมีการควบคุมการรับ–ส่งข้อมูลด้วย sequence number และ acknowledgement

d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values.

SOURCE PORT : 1027

DESTINATION PORT) : 80

SEQUENCE NUMBER : 1

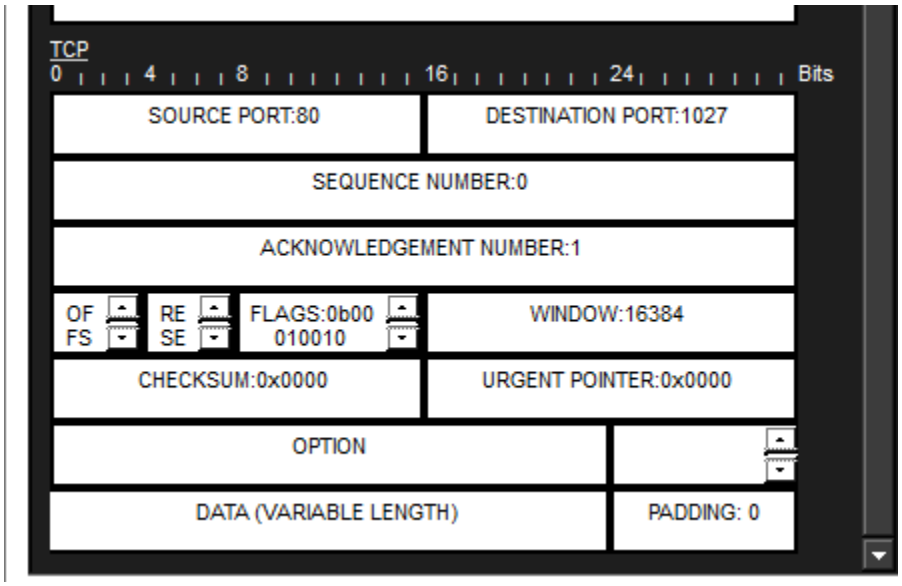
ACKNOWLEDGEMENT NUMBER : 1

Question

Q: What is the value in the flag field?
Ans :flag ที่ถูกตั้งค่าใน PDU นี้คือ RST และ

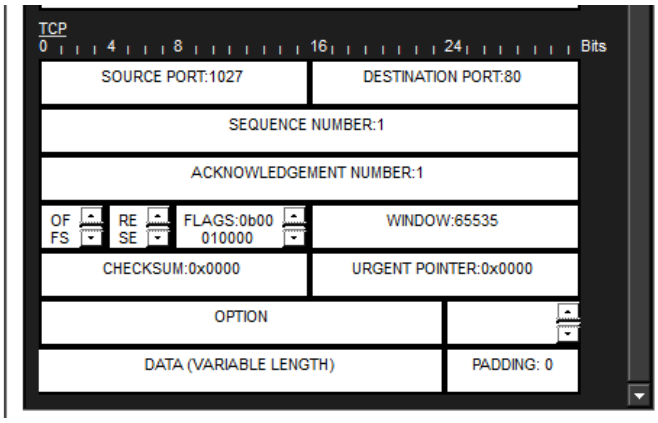
e. Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.

f. Click the PDU envelope and select **Inbound PDU Details**.



How are the port and sequence numbers different than before?

g. Click the **Outbound PDU Details** tab.

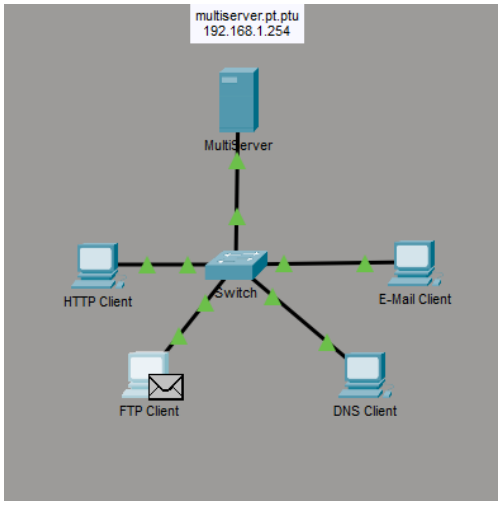


How are the port and sequence numbers different from the previous results?

Ans: หมายเลข SRC PORT และ DEST PORT ถูกสลับตำแหน่งกัน และค่า SEQUENCE NUM และ ACK

NUM เปลี่ยนไป เพื่อยืนยันการรับข้อมูล แสดงถึงการตอบกลับจากฝั่งเซิร์ฟเวอร์ในขั้นตอนของ TCP

h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.



i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section.

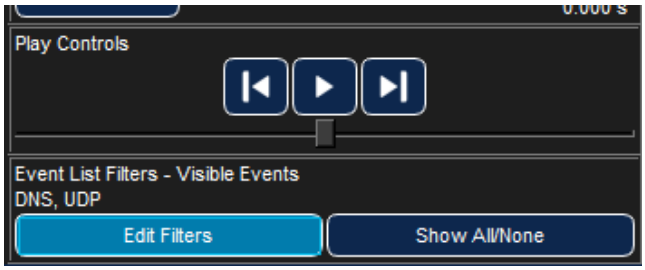
What is the message from the server?

j. Click Reset Simulation.

Step 3: Examine DNS traffic as the clients communicate with the server.

a. Repeat the steps in Part 1 to create DNS traffic.

b. In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.



c. Click the PDU envelope to open it.

d. Look at the OSI Model details for the outbound PDU.

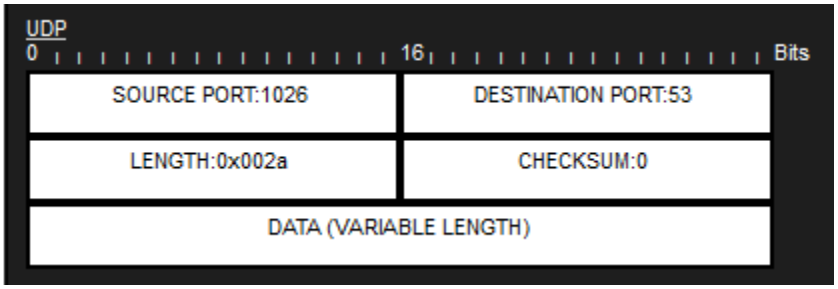
What is the Layer 4 protocol?

Ans : UDP

Are these communications considered to be reliable?

Ans ไม่ เพราะ UDP ไม่มี handshake ไม่มี ACK

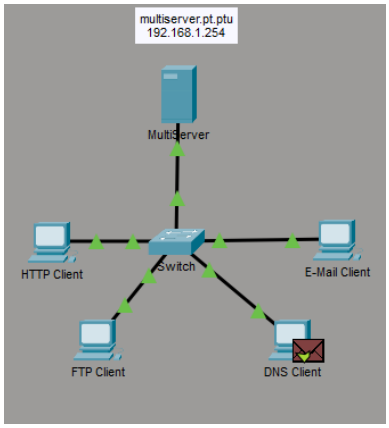
e. Open the Outbound PDU Details tab and find the UDP section of the PDU formats. Record the **SRC PORT** and **DEST PORT** values.



Why are there no sequence and acknowledgement numbers?

Ans: UDP เป็นโปรโตคอลที่ไม่ต้องสร้างการเชื่อมต่อล่วงหน้า และไม่ใช้หมายเลขลำดับหรือหมายเลขยืนยัน

f. Close the **PDU** and click **Capture/Forward** until a PDU with a check mark returns to the **DNS Client**.



g. Click the PDU envelope and select **Inbound PDU Details**.

How are the port and sequence numbers different than before?

Ans : พอร์ตต้นทางและปลายทางสลับกัน

What is the last section of the **PDU** called?

Ans : Domain Name System (DNS)

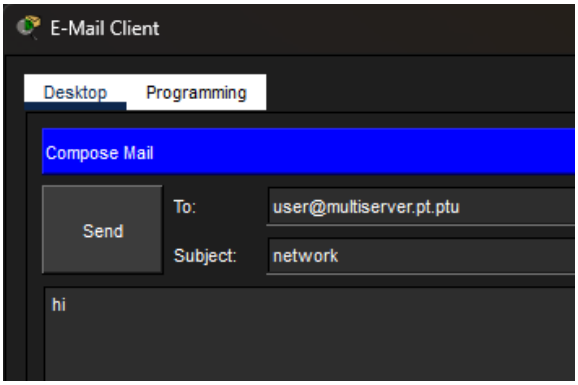
What is the IP address for the name **multiserver.pt.ptu**?

Ans : Address: 192.168.1.254

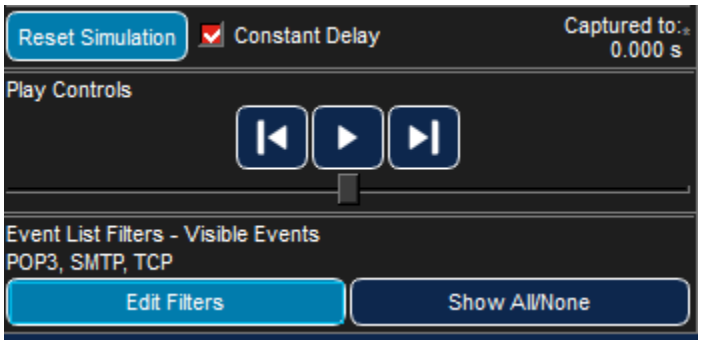
h. Click Reset Simulation.

Step 4: Examine email traffic as the clients communicate with the server.

a. Repeat the steps in Part 1 to send an email to **user@multiserver.pt.ptu**.

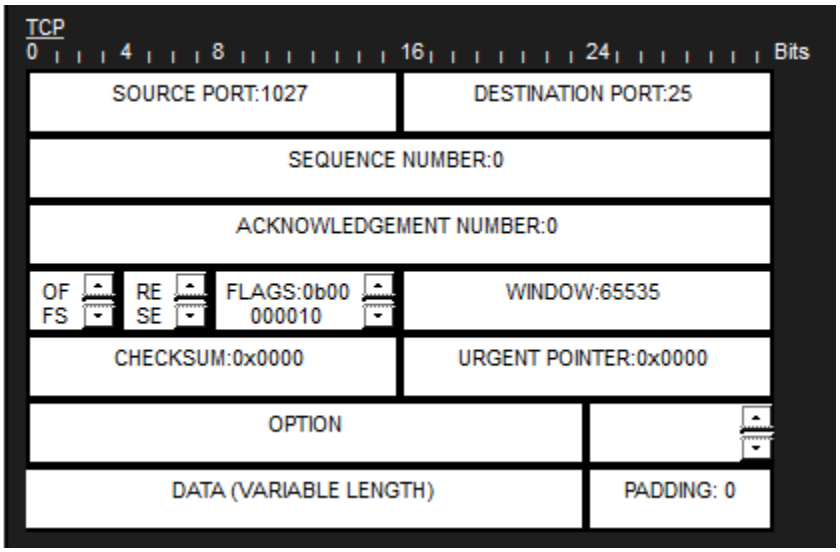


b. In the Simulation Panel, change **Edit Filters** to display only **POP3, SMTP** and **TCP**.



c. Click the first PDU envelope to open it.

d. Click the **Outbound PDU Details** tab and scroll down to the last section.



What transport layer protocol does email traffic use?

Ans : TCP

Are these communications considered to be reliable?

Ans: เชื่อถือได้ เพราะ TCP มี acknowledgement ,retransmission,sequence tracking

e. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is the flag field value?

SRC PORT: random 1027

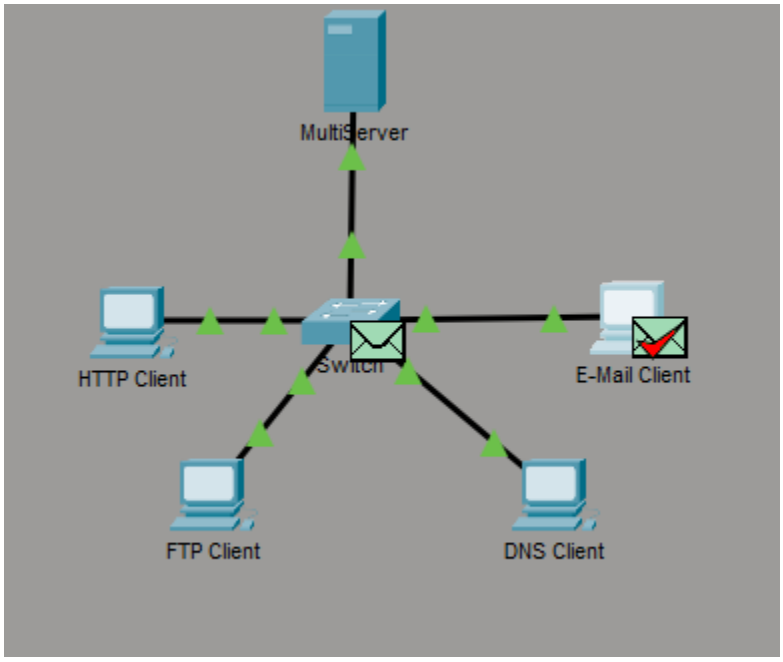
DEST PORT: 25

SEQUENCE NUM: 0

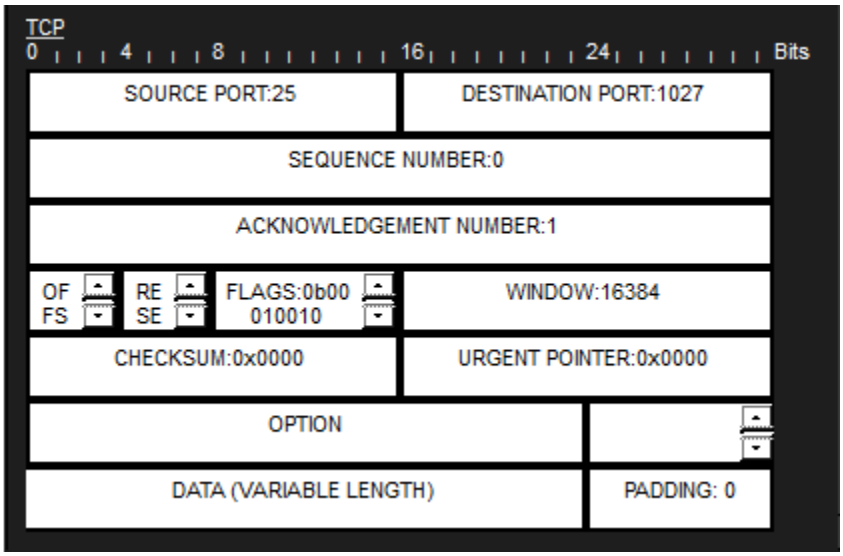
ACK NUM: 0

FLAG: SYN

f. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.



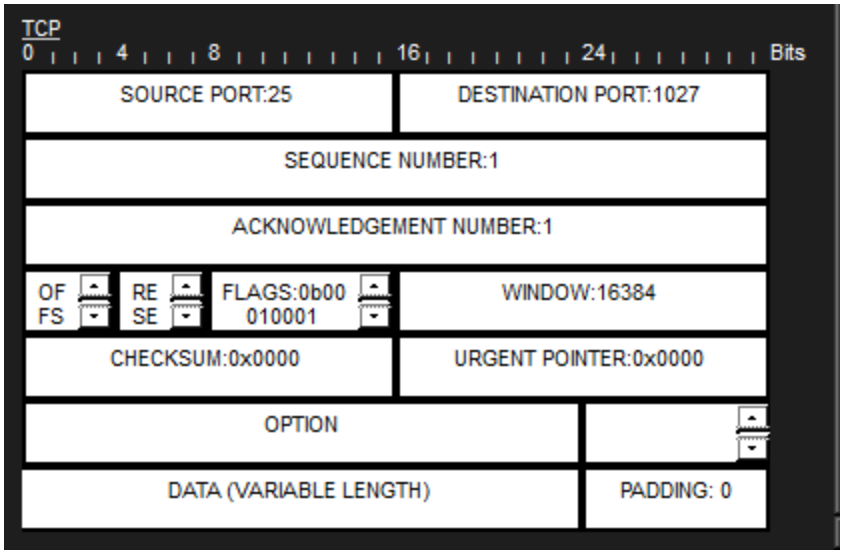
g. Click the TCP PDU envelope and select **Inbound PDU Details**.



How are the port and sequence numbers different than before?

Ans: พอร์ตต้นทางและปลายทางสลับกันและหมายเลขการรับทราบเพิ่มขึ้น

h. Click the **Outbound PDU Details** tab.



How are the port and sequence numbers different from the previous two results?

Ans : หมายเลขลำดับจะเพิ่มขึ้นเมื่อมีการส่งข้อมูลและหมายเลขยืนยันจะยืนยันว่าได้รับข้อมูลแล้ว

i. There is a second **PDU** of a different color that **E-Mail Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details**.

How are the port and sequence numbers different from the previous two **PDU**s?

Ans: หมายเลขลำดับยังคงเพิ่มขึ้นเรื่อยๆซึ่งบ่งชี้ว่ามีการส่งข้อมูลอย่างต่อเนื่อง

What email protocol is associated with TCP port 25?

Ans: SMTP

What protocol is associated with TCP port 110?

Ans: POP3